# Appendix: Data Cleaning

Anomaly Detection on Heterogeneous Graphs for the Italian Public Administration

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## Contents

# 1 Data Cleaning

This report describes the data preprocessing pipeline developed within the project "Anomaly Detection on Heterogeneous Graphs for the Italian Public Administration", whose objective is to construct a heterogeneous graph representing the network of Italian public procurement procedures and to apply anomaly detection techniques on such structure. In particular, preprocessing constitutes the fundamental preliminary phase of the project, aimed at cleaning, harmonizing, and transforming the administrative databases provided by the National Anti-Corruption Authority (ANAC), in order to obtain coherent and relationally consistent numerical datasets, ready for graph construction and for the training of graph-based anomaly detection models.

## 1.1 Objective

- data cleaning;
- construction of coherent datasets for the creation of node attributes (SA, CIG, AGZ, AG) within the heterogeneous graph representing the network of public procurement procedures in the Italian Public Administration;
- encoding of variables into numerical format, suitable for tensor representations required by the proposed algorithm.

## 1.2 Input files

All the following files were downloaded from the Open Data Portal of the National Anti-Corruption Authority [1] on February 10, 2026.

- `stazioni_appaltanti_csv.csv`, containing registry information on contracting authorities;
- `cig_csv_2022.csv`, `cig_csv_2023.csv`, `cig_csv_2024.csv`, and `cig_csv_2025.csv`, containing public procurement tender data;
- `aggiudicazioni_csv.csv`, containing the outcomes of procurement procedures;
- `aggiudicatari_csv.csv`, containing information on winning economic operators.

## 1.3 Output files

- `stazioni_appaltanti_v2.parquet`, `CIG_v2.parquet`, `aggiudicazioni_v2.parquet`, and `aggiudicatari_v2.parquet`, containing the final numerical features for each node type;
- mapping files `SA_id.json`, `CIG_id.json`, `AGZ_id.json`, and `AG_id.json`, preserving the original identifying information;
- auxiliary files `SA_ca.json`, `CIG_cig.json`, `AG_cf.json`, `AGZ_idag.json`, and `aggiudicatari_links.parquet`, used to reconstruct the relationships among entities during graph construction;

- the file `PA_graph.pt`, containing the final heterogeneous graph in PyTorch Geometric format, complete with nodes, features, and relational edges.

## 1.4 Preprocessing

### 1.4.1 Contracting Authorities

For the contracting authorities dataset, the file is read from CSV format, duplicates are removed, and the main variables are processed.

- `codice_fiscale` and `denominazione` are converted into string format;
- `codice_ausa` is normalized using the function `fix_codice_ausa`;
- the province code is cleaned by removing the prefix `IT-` and converting it into the corresponding provincial abbreviation. Starting from the province information, the region is then manually reconstructed through a mapping dictionary from provincial abbreviation to region. Subsequently, the variable `regione` is expanded through one-hot encoding;
- the `soggetto_estero` flag is completed: missing values are set to 0, but if the province is `EE` or `99`, the record is forced to be classified as foreign;
- boolean or quasi-boolean flags such as `flag_inHouse` and `flag_partecipata` are processed by converting them into `int8`;
- the categorical variables `stato` and `natura_giuridica_codice` are transformed using dummy variables;
- the following columns are discarded because they are considered unnecessary for the final numerical representation, either due to redundancy or excessive corruption: `data_inizio`, `data_fine`, `partita_iva`, `natura_giuridica_descrizione`, `citta_nome`, `indirizzo_odonimo`, `cap`, `citta_codice`.

The resulting dataset is saved as `stazioni_appaltanti.parquet`.

### 1.4.2 CIG

For the CIG dataset, the script separately reads the annual files for 2022, 2023, 2024, and 2025, concatenates them vertically, and removes duplicates. It then performs a conceptually important filtering step: all tenders with outcomes incompatible with the construction of the final graph are removed, including deserted tenders, cancelled procedures, non-awarded contracts, procedures still in progress, or procedures closed without a selected winner. The variables are then processed as follows:

- the columns `ESITO` and `COD_ESITO` are removed;
- `cig` is converted into string format;

- cig_accordo_quadro is summarized into a binary variable cig_aq, indicating the presence or absence of a framework agreement reference;
- importo_complessivo_gara, n_lotti_componenti, importo_lotto, and IMPORTO_SICUREZZA are standardized and enriched with missing value indicators;
- the variable oggetto_principale_contratto is transformed through one-hot encoding;
- stato is binarized into active/non-active status;
- settore is binarized into ordinary sectors versus all remaining categories;
- sezione_regionale, FUNZIONI_DELEGATE, and IPOTESI_COLLEGAMENTO are recoded using dummy variables after a harmonization phase of textual values;
- flag_prevalente, FLAG_URGENZA, FLAG_DELEGA, and FLAG_PNRR_PNC are converted into binary numerical format;
- the relational identifying fields codice_ausa, cf_amministrazione_appaltante, CF_SA_DELEGATA, CF_SA_DELEGANTE, and CIG_COLLEGAMENTO are cleaned as strings, but not immediately removed, since they will be required during the second phase of alignment and identifier construction;
- data_pubblicazione is converted into datetime format. The variable will later be used for the numerical encoding of other date variables and then removed;
- data_scadenza_offerta is converted into datetime format and then replaced with the number of days between the publication date of the tender and the offer deadline, accompanied by a missing-value flag and completed through zero imputation. This difference in days is subsequently standardized;
- the following columns are discarded because they are considered unnecessary for the final numerical representation, either due to redundancy or excessive corruption: oggetto_gara, oggetto_lotto, cpv, codice_cpv, descrizione_cpv, modalita_realizzazione, motivazione_urgenza, dettaglio_urgenza, descrizione_stato, denominazione_amministrazione_appaltante, denominazione_centro_costo, descrizione_settore, descrizione_tipo_contratto, descrizione_procedura, descrizione_sceltacontraente.

The resulting dataset is saved as CIG.parquet.

### 1.4.3 Awards

For the awards dataset, the file is read from CSV format and immediately subjected to winner-level deduplication. The variables are then processed as follows:

- the column cig is converted into string format;

- the variable esito is normalized and then used as a selective filter: the dataset is restricted only to records with outcome "AWARDED", thereby removing cancelled, deserted, or otherwise incomplete procurement procedures; after filtering, the esito column itself is removed. This is consistent with the project logic, which considers only awarded tenders and complete relational chains;

- flag_subappalto, asta_elettronica, FLAG_PROC_ACCELERATA, and FLAG_SCOMPUTO are converted into integers, with zero imputation where necessary;

- the dates data_comunicazione_esito and data_aggiudicazione_definitiva are first converted into datetime format using robust parsing, leaving NaT values for non-interpretable cases; dates earlier than January 1, 2000 are considered spurious and converted into missing values; subsequently, each date is replaced with the difference in days relative to the data_pubblicazione of the corresponding tender. These durations are also standardized using standardize_col, enriched with missing-value indicators, and completed through zero imputation;

- the numerical variables importo_aggiudicazione and ribasso_aggiudicazione are standardized using standardize_col. For each of them, the script creates a corresponding missing-value indicator (*_NA) and then replaces missing values with zero. In the case of discount percentages, additional cleaning is applied: negative values are converted into absolute values, while discounts exceeding 100% are marked as missing;

- the following columns are discarded because they are considered unnecessary for the final numerical representation, either due to redundancy or excessive corruption: numero_offerte_ammesse, numero_offerte_escluse, numero_offerte_presentate, numero_imprese_offerenti, flag_progettazione_esterna, flag_progettazione_interna, esito, modalita_riaggiudicazione, motivazione_riaggiudicazione, descrizione_esito.

The resulting dataset is saved as aggiudicazioni.parquet.

### 1.4.4 Awardees

For the awardees dataset, the script loads the CSV file, removes duplicates, and then processes the variables as follows:

- the variable cig is converted into string format;

- an important manual correction step is performed on the codice_fiscale field: for some specific entities, such as "NOVARTIS FARMA SPA", "SETTEN SRL", "SOLETO SPA", and other isolated cases, the tax identification number is manually completed or corrected directly within the script (this procedure could be extended to many more specific cases, although it requires considerable time and attention). After these corrections, codice_fiscale is cleaned from spaces and empty strings; records still lacking a tax identification number are completely removed;

- `denominazione` is cleaned from spaces and empty values;
- the variable `tipo_soggetto` is harmonized by grouping long or inconsistent descriptions into a more compact set of standardized categories, such as `IMPRESA_SINGOLA`, `CONSORZIO`, `ATI`, `GEIE`, `DITTA_INDIVIDUALE`, `STAZIONE_APPALTANTE`, `MONOSOGGETTIVO`, and others. The recoded categories are then transformed into dummy variables;
- `ruolo` is completely removed.

The resulting dataset is saved as `aggiudicatari.parquet`.

## 1.5 Database Filtering

A relational alignment among the four tables is performed through key intersection rather than through traditional joins.

First, only the pairs ($id\_aggiudicazione, cig$) present in both `aggiudicazioni` and `aggiudicatari` are retained, ensuring the consistency of the AGZ → AG relationship. During this phase, a manual correction of anomalies within the awards dataset is also performed, removing inconsistent records and correcting specific cases identified through manual inspection.

Subsequently, only the CIG codes also present in the `aggiudicazioni` table are retained, ensuring the consistency of the CIG → AGZ relationship and consequently filtering the awardees dataset as well.

Finally, only the `codice_ausa` values shared between contracting authorities and CIG records are preserved, enforcing the consistency of the SA → CIG relationship.

A final refinement iteratively reconstructs the set of valid CIG codes and filters all tables once again, retaining exclusively the entities for which it is possible to reconstruct the entire relational chain SA → CIG → AGZ → AG.

## 1.6 Identifiers

For the contracting authorities, a progressive numerical identifier is created. Then, the dictionary `SA_id` is constructed, associating each identifier with the textual information `codice_fiscale`, `denominazione`, and `codice_ausa`, together with the dictionary `SA_ca`, which maps each `codice_ausa` to the corresponding numerical identifier. After constructing these two dictionaries, the three identifying columns are removed from the final numerical matrix, and the dataset is saved as `stazioni_appaltanti_v2.parquet`. The two dictionaries are saved as `SA_id.json` and `SA_ca.json`.

- `SA_id` : $id \rightarrow codice\_fiscale, codice\_ausa$
- `SA_ca` : $codice\_ausa \rightarrow id$

For the CIG dataset, a numerical identifier is created and two dictionaries are constructed: `CIG_id`, which stores for each identifier the values `cig`, `codice_ausa`, `id_centro_costo`, `CIG_COLLEGAMENTO`, `CF_SA_DELEGATA`, and `CF_SA_DELEGANTE`; and `CIG_cig`, which maps the textual identifier `cig` to

the numerical node identifier. Once these dictionaries have been created, both the main textual keys and other unnecessary identifying variables such as
`cf_amministrazione_appaltante`, `denominazione_amministrazione_appaltante`, `id_centro_costo`, and `denominazione_centro_costo` are removed from the final dataset. The resulting dataset is saved as
`CIG_v2.parquet`, together with the corresponding files `CIG_id.json` and `CIG_cig.json`.

- `CIG_id` : $id \rightarrow cig, codice\_ausa, id\_centro\_costo, CIG\_COLLEGAMENTO, CF\_SA\_DELGATA, CF\_SA\_DELEGANTE$
- `CIG_cig` : $cig \rightarrow id$

For the awards dataset, a progressive numerical identifier is created and two dictionaries are produced: `AGZ_id`, which associates each numerical identifier with the information pair $\{cig, id\_aggiudicazione\}$, and `AGZ_idag`, which uses the composite key `"{id_aggiudicazione}_{cig}"` to retrieve the node identifier. After constructing the dictionaries, the columns `cig` and `id_aggiudicazione` are removed from the feature matrix, and the final dataset is saved as `aggiudicazioni_v2.parquet`, together with the two JSON files.

- `AGZ_id` : $id \rightarrow (cig, id\_aggiudicazione)$
- `AGZ_idag` : $(cig, id\_aggiudicazione) \rightarrow id$

For the awardees dataset, an auxiliary object called `aggiudicatari_links` is first constructed, containing the triples $(codice\_fiscale, cig, id\_aggiudicazione)$ after duplicate removal. This file preserves the granularity of the relationships among the economic operator, the tender, and the award. Immediately afterward, however, the main `aggiudicatari` dataset is collapsed at the economic operator level: the variables `codice_fiscale`, `denominazione`, and all dummy columns beginning with `tipo_` are selected and aggregated by `codice_fiscale`, using the first available denomination and the maximum value over the type dummy variables. In this way, multiple occurrences of the same operator across different procurement procedures are merged into a single AG node. A numerical identifier is then assigned, the dictionaries `AG_id` and `AG_cf` are created, the textual identifying columns are removed, and the final versions `aggiudicatari_v2.parquet`, `AG_id.json`, `AG_cf.json`, together with the relational file `aggiudicatari_links.parquet`, are saved.

The original awardees table may contain the same company repeated multiple times, once for each awarded contract. The script therefore separates two logical levels: on one side, it constructs the AG node as a unique economic operator identified by the tax identification number; on the other side, it preserves in `aggiudicatari_links` the trace of that operator's participation in individual awards. This makes it possible to construct a graph where companies are not artificially duplicated, while the AGZ $\rightarrow$ AG relationships remain fully recoverable.

- `AG_id` : $id \rightarrow codice\_fiscale$
- `AG_cf` : $codice\_fiscale \rightarrow id$

## 1.7 Summary

Overall, the preprocessing pipeline simultaneously achieves five objectives:

- cleaning and harmonizing heterogeneous variables originating from highly noisy administrative sources;
- transforming as many fields as possible into numerical or binary format, making them usable as node features;
- explicitly representing missingness information through dedicated indicator columns;
- filtering the global dataset until only relationally compatible entities with the graph structure remain;
- systematically separating the numerical feature component from the identifying component, which is preserved in external dictionaries in order to enable node interpretation after model training.

From a practical perspective, the final output of the preprocessing phase consists of a collection of `*_v2.parquet` tables already prepared for conversion into numerical tensors, together with a set of JSON dictionaries allowing the reconstruction of the semantic meaning of nodes and relational keys.

## 1.8 Additional Information

- the dictionary `SCALER_PARAMS`, used to store the means and standard deviations employed during standardization procedures; once the separate cleaning of the four datasets is completed, the script saves the `SCALER_PARAMS` dictionary into the file `models/scaler_params.json`, in order to preserve the standardization statistics used for the numerical columns.
- the function `fix_codice_ausa`, which normalizes the `codice_ausa` field by removing spaces, possible `.0` suffixes, leading zeros, and empty strings.

## 1.9 Structure of Input Tables

### 1.9.1 Contracting Authority

| Field | Description | Type |
|---|---|---|
| codice_fiscale | Tax Identification Number of the contracting authority. | string |
| partita_iva | VAT number of the contracting authority. | string |
| denominazione | Official name of the entity. | string |
| codice_ausa | Unique identifier code in the ANAC registry. | string |
| natura_giuridica_codice | Code representing the legal form of the entity. | string |
| natura_giuridica_descrizione | Description of the legal form (e.g. Municipality, Local Health Authority). | string |
| soggetto_estero | Flag indicating whether the entity is based abroad. | boolean |
| provincia_nome | Name of the province of the legal headquarters. | string |

| Field | Description | Type |
|---|---|---|
| citta_nome | Municipality of the legal headquarters. | string |
| indirizzo_odonimo | Full address of the headquarters. | string |
| cap | Postal code. | string |
| flag_inHouse | Indicates whether the entity is an “in-house” company. | boolean |
| flag_partecipata | Indicates whether the company is publicly owned or participated by public entities. | boolean |
| Stato | Current status of the record (active, terminated, etc.). | string |
| data_inizio | Start date of validity of the information. | date |

### 1.9.2 Awards

| Field | Description | Type |
|---|---|---|
| cig | Identification code of the award. | string |
| data_aggiudicazione_definitiva | Award date, as specified in the official report. | string |
| esito | Description of the procurement outcome (Awarded, Cancelled. . . ). | string |
| criterio_aggiudicazione | Award criterion adopted (most economically advantageous offer, lowest price, etc.). | string |
| data_comunicazione_esito | Date of communication of the award outcome. | string |
| numero_offerte_ammesse | Number of admitted offers. | bigint |
| numero_offerte_escluse | Number of excluded offers. | bigint |
| importo_aggiudicazione | Award amount. | double |
| ribasso_aggiudicazione | Discount percentage offered by the awarded economic operator. | double |
| num_imprese_offerenti | Number of companies that submitted an offer. | bigint |
| flag_subappalto | Indicator of subcontracting possibility. | boolean |
| id_aggiudicazione | Unique award identifier. | bigint |
| cod_esito | Identification code of the procurement outcome. | bigint |
| num_imprese_richiedenti | Number of companies that requested to be invited. | bigint |
| asta_elettronica | Indicator of the use of electronic auctions. | bigint |
| num_imprese_invitate | Number of companies invited to the tender. | bigint |
| massimo_ribasso | Maximum discount percentage offered. | double |
| minimo_ribasso | Minimum discount percentage offered in the tender. | double |
| FLAG_SCOMPUTO | Indicates whether the contract concerns urbanization works in lieu of payment. | boolean |
| COD_PRESTAZIONI_COMPRESE | Code indicating whether the contract concerns execution only or also includes design activities. | bigint |
| PRESTAZIONI_COMPRESE | Description indicating whether the contract concerns execution only or also includes design activities. | string |
| CIG_PROG_ESTERNA | CIG associated with the external design assignment. | string |
| DATA_INCARICO_PROG | Date of assignment of the external design activity. | date |
| DATA_CONS_PROG | Project delivery date in the case of external design activities. | date |
| COD_MODO_RIAGGIUDICAZIONE | Code identifying the re-award procedure method. | bigint |
| MODO_RIAGGIUDICAZIONE | Description of the re-award procedure method. | string |
| FLAG_PROC_ACCELERATA | Indicates whether an accelerated emergency procedure was adopted. | boolean |
| N_MANIF_INTERESSE | Number of entities that submitted an expression of interest (restricted procedures). | bigint |

### 1.9.3 Awardees

| Field | Description | Type |
|---|---|---|
| cig | Tender Identification Code assigned by the Authority to uniquely track procurement procedures and contracts at the national level (assigned at lot level). | string |
| ruolo | In the case where the awardee is a consortium or grouping of companies, indicates the role of the company within the group. | string |
| codice_fiscale | Tax Identification Number of the awardee. | string |
| denominazione | Name of the awardee. | string |
| tipo_soggetto | Specifies whether the awardee is a single company or a consortium/grouping. | string |
| id_aggiudicazione | Unique award identifier. Connects the awardee to the related data record and distinguishes among multiple awardees for the same CIG. | bigint |

### 1.9.4 CIG Tender

| Field | Description | Type |
|---|---|---|
| cig | Tender Identification Code (CIG). | string |
| cig_accordo_quadro | CIG of the related framework agreement (if existing). | string |
| numero_gara | Identification number of the procurement procedure (may include multiple lots/CIGs). | string |
| oggetto_gara | Tender subject – description of goods/services. | string |
| importo_complessivo_gara | Total amount of the procurement procedure. | double |
| n_lotti_componenti | Number of lots composing the contract. | bigint |
| oggetto_lotto | Lot description. | string |
| importo_lotto | Lot amount. | double |
| oggetto_principale_contratto | Works, Services, or Supplies. | string |
| stato | Lifecycle status of the CIG. | string |
| settore | Ordinary sectors (Directive 24 EU) or Special sectors (Directive 25 EU). | string |
| luogo_istat | ISTAT location code. | string |
| provincia | Province abbreviation. | string |
| data_pubblicazione | Tender publication date. | Date |
| data_scadenza_offerta | Deadline for offer submission. | date |
| cod_tipo_scelta_contraente | Procurement procedure selection code. | string |
| tipo_scelta_contraente | Description of the procurement procedure type. | string |
| cod_modalita_realizzazione | Execution modality code. | bigint |
| modalità_realizzazione | Description of the execution modality. | string |
| codice_ausa | ANAC identification code of the contracting authority. | string |
| cf_amministrazione_appaltante | Tax identification number of the contracting authority. | string |
| denominazione_amministrazione_appaltante | Name of the contracting authority. | string |
| sezione_regionale | Regional section responsible for data submission. | string |
| id_centro_costo | Cost center identifier. | string |
| denominazione_centro_costo | Name of the cost center. | string |
| anno_pubblicazione | Publication year. | string |
| mese_pubblicazione | Publication month. | string |
| cod_cpv | Commodity category code (CPV vocabulary). | string |
| descrizione_cpv | Commodity category description. | string |
| flag_prevalente | Indicates whether the commodity category is the main one. | bigint |
| MOTIVO_CANCELLAZIONE | Reason for cancellation (if status = CANCELLED). | string |
| DATA_ULTIMO_PERFEZIONAMENTO | Date of the latest completion/update. | date |
| FLAG_URGENZA | Extreme urgency / emergency procedure indicator. | boolean |
| IMPORTO_SICUREZZA | Safety-related amount (not subject to discount). | double |
| Flag_PNRR_PNC | Contract financed through PNRR and/or PNC resources. | boolean |

## References

[1] National Anti-Corruption Authority (ANAC). *ANAC Open Data Portal*. Available at: https://dati.anticorruzione.it/opendata/ (accessed: February 2026).